

ORIGINAL RESEARCH

CORONARY

Impact of Myocardial Bridges on Long-Term Outcomes After Left Anterior Descending Chronic Total Occlusion Recanalization



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ABSTRACT

BACKGROUND A myocardial bridge (MB) is frequently observed after recanalization of a left anterior descending coronary artery (LAD) chronic total occlusion (CTO); however, the long-term prognostic impact of MBs in this specific setting remains controversial.

OBJECTIVES The aim of this study was to determine the impact of MBs on long-term outcomes in patients undergoing percutaneous coronary intervention of LAD CTOs.

METHODS Consecutive patients with LAD CTOs who underwent intravascular ultrasound-guided percutaneous coronary intervention from January 2019 to December 2021 were retrospectively analyzed. The primary endpoint was the 5-year rate of major adverse cardiac events (MACE), a composite of cardiac death, spontaneous myocardial infarction, and ischemia-driven revascularization.

RESULTS Among 685 patients analyzed, MBs were identified in 393 (57.4%) on intravascular ultrasound; stent extension into the bridged segment occurred in 44.3% of these cases (174 of 393). After a median 5-year follow-up, patients with MBs had a higher incidence of MACE compared with those without MBs (16.1% vs 8.9%; $P = 0.006$), driven primarily by more ischemia-driven revascularization in the MB group (15.1% vs 7.9%; $P = 0.004$). Stent extension into the MB was associated with a higher rate of MACE compared with sparing the MB (21.4% vs 11.9%; $P = 0.0017$). MACE rates did not differ between patients with MBs spared from stenting and those without MBs (11.9% vs 8.9%; $P = 0.25$).

CONCLUSIONS MBs are prevalent after LAD CTO recanalization. Although their presence is associated with adverse outcomes, the risk is driven mainly by stent extension into the bridged segment. (JACC Cardiovasc Interv. 2026;19:2220-2230) © 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

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With advances in algorithms, techniques, and devices, success rates for chronic total occlusion (CTO) percutaneous coronary intervention (PCI) have increased to nearly 90%; however, maintaining long-term patency and preventing late adverse events remain major challenges.^{1,2} A myocardial bridge (MB) is the most common congenital coronary variation, characterized by a muscle band overlying the epicardial coronary artery. Prior autopsy studies have described MBs as a “double-edged sword,” protecting against atherosclerosis within the bridges but accelerating plaque formation proximally.³ This phenomenon may account for recent findings of a significantly higher prevalence of MBs observed following left anterior descending coronary artery (LAD) CTO recanalization.^{4,5} The systolic compression within the bridged segment can obscure anatomical boundaries during angiography, thereby increasing the risk for unintended stent extension into the MB. As this procedure is potentially associated with adverse outcomes,^{4,6} increasing attention has focused on the impact of concurrent MBs on long-term prognosis. Although recent data from Yamamoto et al⁴ highlight the prevalence of MBs in LAD CTOs, that study was limited by a small sample size and short follow-up duration. The impact of specific morphologies, including the thickness or length of MBs, on stent failure remains undercharacterized in large populations. Therefore, we conducted this large-scale study of consecutive patients undergoing intravascular ultrasound (IVUS)-guided recanalization of LAD CTOs to investigate the association between MB presence and long-term outcomes and to evaluate the prognostic impact of stent extension into the bridged segment.

METHODS

STUDY POPULATION. We retrospectively analyzed consecutive patients with LAD CTOs who underwent PCI at Fuwai Hospital from January 2019 to December 2021. A CTO was defined as a 100% coronary occlusion with a TIMI flow grade of 0 and a clinically estimated occlusion duration of more than 3 months, as determined by an independent imaging core laboratory. The median follow-up duration was 5 years.

The inclusion criteria were: 1) age $\geq$ 18 years; 2) successful recanalization of the LAD CTO, defined as successful revascularization with $<30\%$ residual diameter stenosis in the treated segment and TIMI grade 3 antegrade flow⁷; and 3) available baseline IVUS images (after wire crossing) and final IVUS

images (after stent implantation). The exclusion criteria were severe valvular or major vascular disease, cardiac transplantation, and severe comorbidities such as severe hepatic or renal dysfunction, hyperthyroidism, active malignancy with extensive metastasis, and bleeding disorders. Patient demographics and procedural details were obtained from electronic medical records. All patients received outpatient or telephone follow-up at 1 and 6 months post-PCI, with subsequent annual follow-up up to 5 years. The study complied with the Declaration of Helsinki and was approved by the institutional review committee of Fuwai Hospital. All eligible patients provided informed consent.

CTO PROCEDURES. All PCI procedures were performed by experienced cardiac interventionalists, with treatment strategies tailored to individual patient characteristics. IVUS examination was performed after successful wire crossing to confirm the wire position, assess lesion morphology, and determine the optimal landing zone and device size. If necessary, further lesion preparation was performed before stent implantation into the reference vessel segment with a plaque burden of $<50\%$.^{2,8} In cases in which MBs were present, stent extension into the bridging segment was generally avoided, except for the presence of significant dissection within the MB or the plaque continuing into the MB.⁴ Procedure success was defined as the restoration of antegrade TIMI grade 3 flow in the target vessel with residual diameter stenosis $<30\%$. Total procedure time was recorded from vascular access establishment to removal of the final catheter.^{9,10}

PERIPROCEDURAL PHARMACOTHERAPY. Patients were prescribed a loading dose of aspirin (300 mg) and either clopidogrel (300 or 600 mg) or ticagrelor (180 mg) before the coronary intervention, unless they received ≥ 6 -day treatment before PCI with aspirin (100 mg/d) and either clopidogrel (75 mg/d) or ticagrelor (90 mg twice daily). Coronary angioplasty and PCI were performed according to current standards and guidelines. The choice of procedure strategy, stent (drug-eluting stent, or bare-metal stent), pre- and postdilation, and the use of intracoronary imaging was left to the operator's discretion on the basis of practice at the time. After the PCI procedure, dual antiplatelet therapy was administered for a minimum of 12 months, with clopidogrel 75 mg/d or ticagrelor 90 mg twice daily on background treatment with aspirin. Any changes to

ABBREVIATIONS AND ACRONYMS

CTO	= chronic total occlusion
DCB	= drug-coated balloon
IPWT	= inverse probability of treatment weighting
IVUS	= intravascular ultrasound
LAD	= left anterior descending
MACE	= major adverse cardiac event(s)
MB	= myocardial bridge
MI	= myocardial infarction
MLA	= minimum luminal area
MSA	= minimum stent area
PCI	= percutaneous coronary intervention

adjunctive pharmacotherapy were made at the discretion of the attending physician.

ANGIOGRAPHIC ANALYSIS. Coronary angiograms and procedural characteristics were retrospectively analyzed by an independent core laboratory (Interventional Cardiovascular Imaging Core Laboratory, National Center for Cardiovascular Diseases, Beijing, China).⁴ The length of CTO lesions was determined according to distal vessel opacification via antegrade or bidirectional contrast injection.⁸ Lesion complexity was graded according to the J-CTO (Multicenter CTO Registry in Japan) score,² while collateral circulation was evaluated using the Rentrop classification system.^{11,12}

IVUS ANALYSIS. Quantitative and qualitative IVUS analyses were independently performed by 2 experienced reviewers (T.Z. and H.R. Su) using a dedicated IVUS analysis platform, in accordance with the latest expert consensus guidelines on intravascular imaging and physiological evaluation.^{13,14} MB was diagnosed by IVUS on the basis of the identification of a “half-moon” sign, characterized by epicardial coronary segment exhibiting systolic compression and surrounded by hypoechoic muscle tissue (**Central Illustration**). The following parameters were measured within the MB segment: minimum luminal area (MLA), plaque burden at the MLA site, maximum MB thickness, total bridge length, systolic and diastolic luminal area at the site of maximal compression, and the degree of lumen compression. CTO length was determined by coregistering IVUS and angiographic images. On IVUS, a CTO was defined as a segment lacking a smooth concave luminal contour, with the cross-sectional area occupied by tissue, compared with the proximal and distal reference segments, which exhibited smooth concave luminal structures. An extraplaque course was identified when the wire traveled outside the plaque but remained within the adventitial layer, characterized by the absence of the 3-layer arterial wall structure. Stent expansion was calculated as the minimum stent area (MSA) divided by the mean luminal area of the proximal and distal reference segments.^{2,4,7}

CLINICAL OUTCOMES. The primary endpoint was 5-year major adverse cardiac events (MACE), a composite of cardiac death, spontaneous myocardial infarction (MI), and ischemia-driven revascularization.¹⁵ Secondary outcomes included individual component of primary outcome, all-cause death, and stroke.^{7,15} Ischemia-driven lesions were defined as restenotic lesions with angiographic diameter

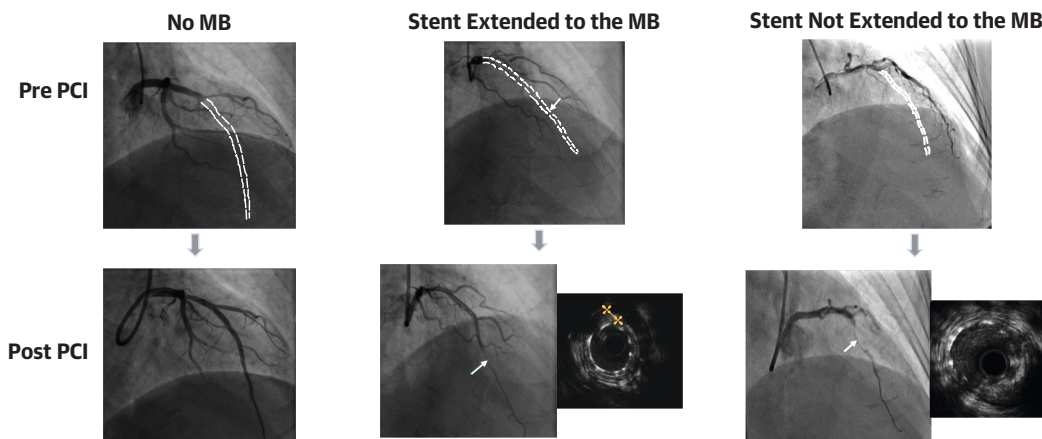
stenosis $\geq 50\%$ with ischemic evidence or $\geq 70\%$ irrespective of the presence of ischemic evidence. All adverse events were adjudicated by an independent clinical events committee.

STATISTICAL ANALYSIS. Continuous variables with a normal distribution are presented as mean $\pm$ SD and were compared using Student's *t*-test. Categorical variables are summarized as frequencies and percentages, and comparisons between groups were made using the chi-square test or the Fisher exact test, as appropriate. Follow-up time was calculated from the date of revascularization, with censoring applied at the time of the last available follow-up. To evaluate the impact of myocardial bridging on long-term prognosis in patients with LAD CTO, the cumulative incidence rates of MACE and secondary outcomes were estimated using the Kaplan-Meier method, and between-group differences were compared using the log-rank test. A multivariable Cox regression model was constructed for adjusted analysis to identify independent predictors of 5-year MACE. The model was adjusted for clinical factors including age, male sex, diabetes mellitus, hypertension, hypercholesterolemia, family history of coronary artery disease, and angina type, as well as lesion factors such as IVUS-measured CTO length and post-PCI MSA, on the basis of clinical relevance and previous studies.^{4,16,17} A prespecified 1-year landmark analysis was conducted to evaluate the late prognostic impact of MB and stenting strategies. Results are reported as HRs with 95% CIs. To mitigate the influence of bias and potential confounders, we adjusted for between-group differences in baseline demographic, angiographic, and IVUS characteristics using inverse probability of treatment weighting (IPTW). The adjusted variables included age, sex, diabetes, J-CTO score, lesion length, calcification, and post-PCI distal reference vessel diameter. Adjusted absolute risk differences and HRs with 95% CIs were calculated using the IPTW-adjusted cohort. Prior to performing the IPTW analysis, we addressed missing data using multiple imputation. Detailed methods for the IPTW analysis are described in the **Supplemental Appendix**. Furthermore, patients were stratified by stent strategy (no MB, stent not extended into MB, stent extended into MB) to assess outcomes, with interaction testing evaluating the relationship between MB presence and stent extension regarding MACE. All statistical tests were 2-sided, and a *P* value < 0.05 was considered to indicate statistical significance. All statistical analyses were performed using R version 4.5.0 (R Foundation for Statistical Computing).

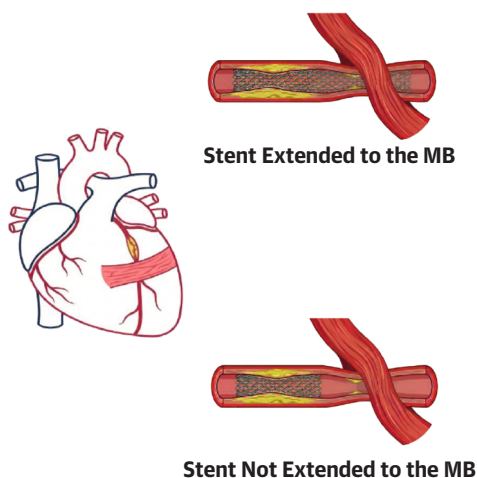
CENTRAL ILLUSTRATION IVUS-Guided PCI in LAD CTO Lesions

Long-Term Outcomes After IVUS-Guided PCI in LAD CTO Lesions With Versus Without Myocardial Bridge, N = 685

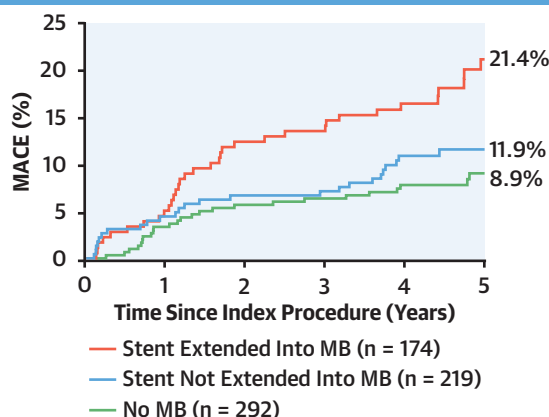
Representative Cases of LAD CTO With an MB



Schematic of LAD CTO With MB



MACE at 5 Years



Stent extended into MB vs Stent not extended into MB

Log-rank $P = 0.017$, HR: 1.82 (95% CI: 1.10-3.01)

Stent extended into MB vs No MB

Log-rank $P < 0.001$, HR: 2.53 (95% CI: 1.53-4.17)

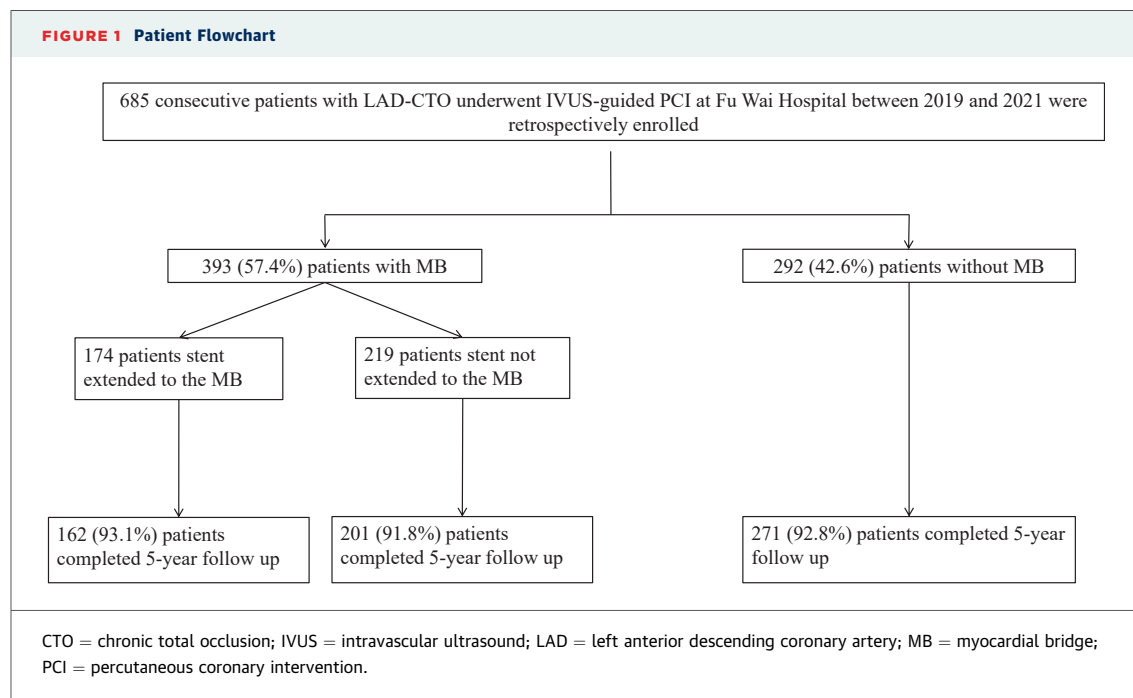
Stent not extended into MB vs No MB

Log-rank $P = 0.25$, HR: 1.38 (95% CI: 0.80-2.38)

- After a median 5-year follow-up, patients with MB had a higher rate of MACE than those without MB (16.1% vs 8.9%, $P = 0.006$).
- Stent extension into the MB was associated with a higher rate of MACE compared with sparing the MB (21.4% vs 11.9%, $P = 0.017$).
- MACE rates did not differ between patients with MBs spared from stenting and those without MB (11.9% vs 8.9%, $P = 0.25$).
- These data support the clinical recommendation to avoid stent placement in the bridged segment whenever feasible.

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Major adverse cardiac events (MACE) are defined as cardiac death, spontaneous myocardial infarction (MI), and ischemia-driven revascularization. The white arrow indicates the site of myocardial bridging, the white dashed line denotes the angiographically identified occlusive segment, and the yellow asterisk marks the location of the myocardial bridge (MB) on intravascular ultrasound. CTO = chronic total occlusion; IVUS = intravascular ultrasound; LAD = left anterior descending coronary artery; PCI = percutaneous coronary intervention.



RESULTS

CLINICAL, ANGIOGRAPHIC, AND PROCEDURAL CHARACTERISTICS. Among 685 patients with successful LAD CTO recanalization, IVUS identified

concomitant MBs in 393 cases (57.4%), whereas the rate of MBs by angiography was 41.6% ([Supplemental Figure 1](#)). Overall, 92.6% of the patients completed the 5-year clinical follow-up ([Figure 1](#)). Patients with and without MBs demonstrated comparable baseline demographic characteristics, although the prevalence of hypertension was significantly lower in the MB group (55.0% vs 64.0%; $P = 0.02$) ([Table 1](#)).

Angiographically, the MB group had significantly shorter lesion length (34.3 ± 13.6 mm vs 37.0 ± 15.0 mm; $P = 0.01$) while other angiographic and procedural characteristics were similar between the groups ([Table 2](#)).

IVUS FINDINGS. Preprocedural IVUS revealed differences between LAD CTO lesions with MBs ($n = 393$) and without MBs ($n = 292$). The MB group had a higher extraplaque tracking rate (7.9% vs 3.1%; $P = 0.008$) and shorter lesion length (40.1 ± 14.5 mm vs 45.7 ± 16.9 mm; $P < 0.001$), whereas CTO length did not differ. No significant between-group differences were observed in calcification or reference vessel area. Post-PCI IVUS showed no differences in MSA or stent expansion between groups. For the MB subgroup, IVUS measured a total MB length of 16.3 ± 8.9 mm, a maximum MB thickness of 0.4 ± 0.2 mm, and an MB compression ratio of $29.0\% \pm 13.2\%$ ([Table 3](#)).

In the subgroup of patients with MBs ($n = 393$), stent extension into the bridged segment occurred

	LAD CTO With an MB (n = 393)	LAD CTO Without an MB (n = 292)	P Value
Age, y	56.2 ± 10.9	56.4 ± 11.1	0.83
Male	328 (83.5)	246 (84.2)	0.78
BMI, kg/m ²	26.3 ± 3.2	26.6 ± 3.1	0.23
Diabetes mellitus	109 (27.7)	80 (27.4)	0.92
Hypertension	216 (55.0)	187 (64.0)	0.02
Hypercholesterolemia	274 (69.7)	207 (70.9)	0.74
Family history of CAD	15 (3.8)	20 (6.8)	0.08
Previous MI	119 (30.3)	99 (33.9)	0.31
Previous PCI	106 (27.0)	81 (27.7)	0.82
Previous CABG	4 (1.0)	4 (1.4)	0.67
Previous stroke	37 (9.4)	33 (11.3)	0.42
Peripheral artery disease	13 (3.3)	12 (4.1)	0.58
Clinical presentation			
Silent ischemia	82 (20.9)	74 (25.4)	0.17
Stable angina	130 (33.1)	90 (30.8)	0.53
Unstable angina	181 (46.0)	128 (43.8)	0.64

Values are mean ± SD or n (%).

BMI = body mass index; CABG = coronary artery bypass grafting; CAD = coronary artery disease; CTO = chronic total occlusion; LAD = left anterior descending coronary artery; MB = myocardial bridge; MI = myocardial infarction; PCI = percutaneous coronary intervention.

in 174 patients (44.3%). Those with stent extension into the MB exhibited longer total MB length (17.4 ± 9.4 mm vs 15.3 ± 8.2 mm; $P = 0.04$), greater MB thickness (0.5 ± 0.3 mm vs 0.4 ± 0.2 mm; $P = 0.03$), larger diastolic and systolic vessel areas at the maximum compression site, and higher plaque burden at the MLA site ($28.0\% \pm 18.6\%$ vs $23.0\% \pm 16.9\%$; $P = 0.01$); no difference was noted in MB compression ratio ($P = 0.88$). Notably, the rate of dissection extending into the MB was also higher in this group (20.1% vs 6.8% ; $P < 0.001$), attributed to the potential need for stent extension to cover dissections that propagated into the MB (Supplemental Table 1). The specific indications for stent extension into the MB are detailed in Supplemental Figure 2, with the leading causes being no available landing zone proximal to the MB for stent implantation (29.3%), flow-limiting dissection extending to the MB (20.1%), and plaque burden $> 50\%$ within the MB segment (20.1%).

CLINICAL OUTCOMES. At 5 years, for patients with stents extending into MBs, the incidence of MACE was significantly higher than for patients with MBs spared from stenting (21.4% vs 11.9% ; $P = 0.017$) and those without MBs (21.4% vs 8.9% ; $P < 0.001$); MACE rates did not differ between patients with MBs spared from stenting and those without MBs (11.9% vs 8.9% ; $P = 0.25$) (Figure 2A, Central Illustration). Consistent findings were observed for ischemia-driven revascularization, with a markedly elevated 5-year incidence in the group with stents extending into MBs group vs not extending into MBs (20.2% vs 11.0% ; $P = 0.015$) and the group without MBs (20.2% vs 7.9% ; $P < 0.001$), with no difference between the latter 2 groups (11.0% vs 7.9% ; log-rank $P = 0.21$) (Figure 2B). Landmark analysis revealed comparable risks across all groups within the first year ($P > 0.05$ for all). However, from 1 to 5 years, stent extension into the MB drove significantly higher risk for late MACE (Figures 2C and 2D). A significant interaction was observed between the presence of an MB and stent extension regarding MACE (P for interaction = 0.006).

Patients with LAD CTO and concomitant MBs had a significantly higher incidence of MACE than those without MBs (16.1% vs 8.9% ; $P = 0.006$) (Supplemental Figure 3A). This difference was driven primarily by higher rates of ischemia-driven revascularization in the MB group (15.1% vs 7.9% ; $P = 0.004$) (Supplemental Figure 3B). No significant differences were observed in all-cause death, cardiac death, spontaneous MI, and stroke (Table 4). Landmark analysis at 1 year showed no significant

TABLE 2 Angiographic and Procedural Characteristics of LAD CTO With vs Without an MB

	LAD CTO With an MB (n = 393)	LAD CTO Without an MB (n = 292)	P Value
Multivessel disease ^a	321 (81.7)	235 (80.5)	0.69
Ostial lesion	98 (24.9)	63 (21.6)	0.31
Lesion length, mm	34.3 ± 13.6	37.0 ± 15.0	0.01
Antegrade procedures	371 (94.4)	280 (95.9)	0.38
Dissection and re-entry	7 (1.8)	2 (0.7)	0.21
Reattempt CTO PCI	4 (1.0)	2 (0.7)	0.64
CTO length, mm	17.6 ± 7.9	17.9 ± 8.1	0.62
CTO length ≥ 20 mm	173 (44.0)	131 (44.9)	0.83
Moderate to severe calcification	67 (17.0)	55 (18.8)	0.55
Lesion bend $> 75^\circ$	38 (9.7)	18 (6.2)	0.10
J-CTO score ≥ 2	101 (25.7)	90 (30.8)	0.14
Post-PCI proximal RVD, mm	3.2 ± 0.4	3.2 ± 0.4	0.67
Post-PCI distal RVD, mm	2.3 ± 0.4	2.3 ± 0.5	0.71
Post-PCI in-segment ^b			
Reference vessel diameter, mm	2.9 ± 0.3	2.9 ± 0.3	0.76
Minimum luminal diameter, mm	2.5 ± 0.4	2.5 ± 0.4	0.47
Diameter stenosis, %	13.8 ± 9.4	12.7 ± 9.4	0.13
Post-PCI TIMI grade 3 flow	374 (95.2)	272 (93.2)	0.26
Post-PCI dissection	35 (8.9)	22 (7.5)	0.52

Values are n (%) or mean $\pm$ SD. ^aMultivessel disease was defined as atherosclerotic stenosis ($\geq 50\%$) involving ≥ 2 major coronary arteries (LAD, left circumflex coronary artery, or right coronary artery). ^bIn-segment includes stent and 5 mm proximal and distal reference from each stent edge.

J-CTO = Multicenter CTO Registry in Japan; RVD = reference vessel diameter; other abbreviations as in Table 1.

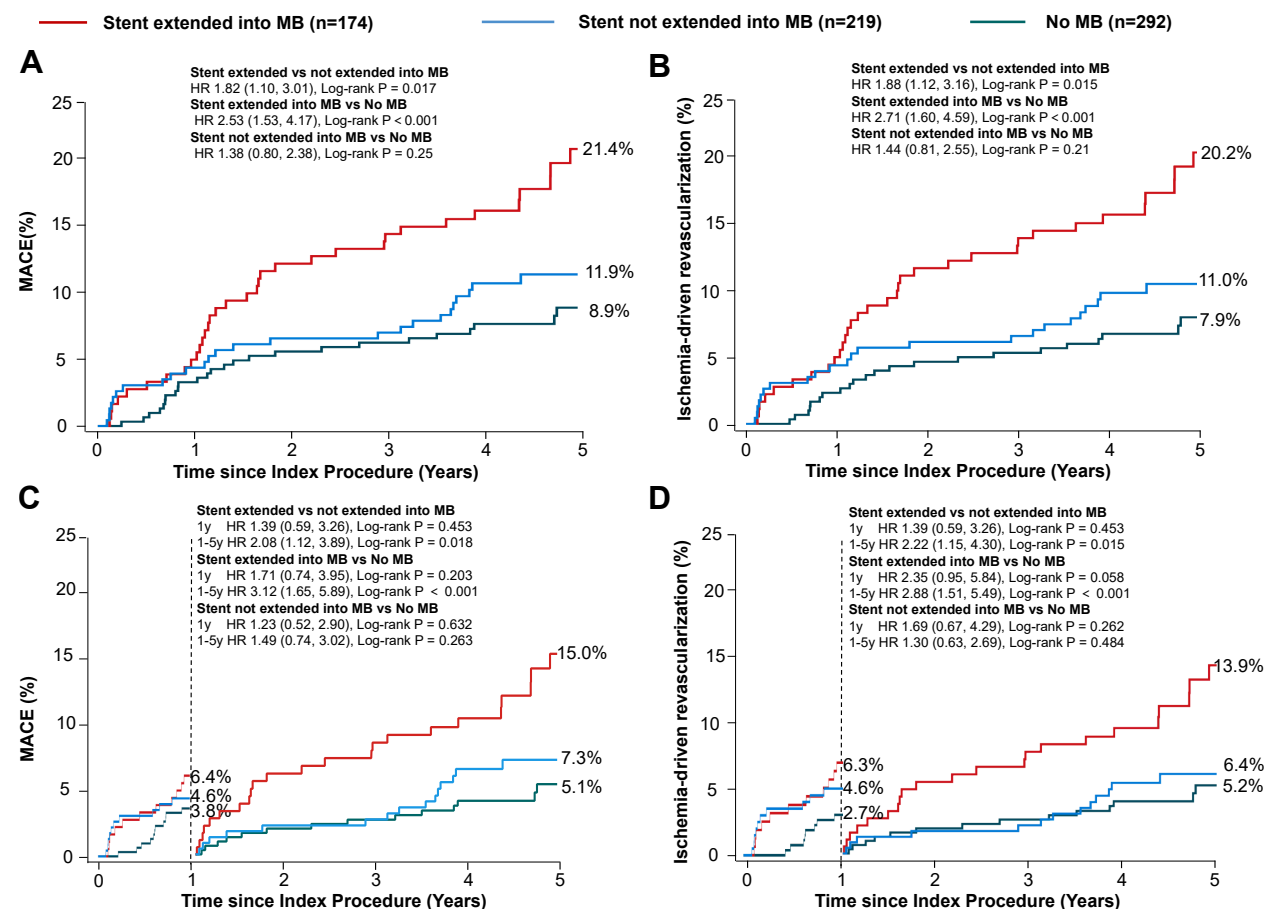
TABLE 3 IVUS Findings of LAD CTO With vs Without an MB

	LAD CTO With an MB (n = 393)	LAD CTO Without an MB (n = 292)	P Value
Preprocedural IVUS lesion characteristics			
Extraplate tracking	31 (7.9)	9 (3.1)	0.008
Lesion length, mm	40.1 ± 14.5	45.7 ± 16.9	<0.001
CTO length, mm	18.4 ± 9.0	18.8 ± 10.3	0.62
CTO length ≥ 20 mm	143 (36.4)	92 (31.5)	0.18
Calcification in CTO lesion	203 (51.7)	145 (49.7)	0.61
Maximum arc of calcium, deg	95.2 ± 110.7	85.9 ± 105.8	0.27
Proximal CTO reference vessel area, mm ²	10.5 ± 2.3	10.2 ± 2.4	0.05
Distal CTO reference vessel area, mm ²	8.2 ± 2.0	8.1 ± 1.9	0.22
MB segment		—	—
Total MB length, mm	16.3 ± 8.9	—	—
Maximum thickness of MB, mm	0.4 ± 0.2	—	—
Diastolic vessel area at maximal compression site, mm ²	4.6 ± 1.8	—	—
Systolic vessel area at maximal compression site, mm ²	3.3 ± 1.5	—	—
MB compression ratio, %	29.0 ± 13.2	—	—
Plaque burden at MLA site, %	25.0 ± 17.9	—	—
Dissection extended into an MB	50 (12.7)	—	—
Post-PCI findings			
MSA, mm ²	4.3 ± 1.2	4.4 ± 1.5	0.36
Stent expansion, %	82.0 ± 13.6	82.0 ± 10.9	0.72
Rate of MSA in the MB, when stented	4.1 ± 1.1	—	—
Dissection	42 (10.7)	25 (8.6)	0.35

Values are n (%) or mean $\pm$ SD.

IVUS = intravascular ultrasound; MLA = minimum luminal area; MSA = minimum stent area; other abbreviations as in Tables 1 and 2.

FIGURE 2 5-Year Rates of MACE and Ischemia-Driven Revascularization Stratified by MB Status and Stenting Strategy



(A) Kaplan-Meier curves for major adverse cardiac events (MACE). (B) Kaplan-Meier curves for ischemia-driven revascularization. (C) Landmark analysis for MACE. (D) Landmark analysis for ischemia-driven revascularization. MACE are defined as a composite of cardiac death, spontaneous myocardial infarction, and ischemia-driven revascularization. MB = myocardial bridge.

TABLE 4 5-Year Clinical Outcomes of LAD CTO With an MB and Without an MB

	Crude				IPTW	
	LAD CTO With an MB (n = 393)	LAD CTO Without an MB (n = 292)	HR (95% CI)	P Value	HR (95% CI)	P Value
MACE	63 (16.1)	26 (8.9)	1.89 (1.20-2.98)	0.006	1.79 (1.13-2.82)	0.01
Cardiac death	4 (1.0)	3 (1.0)	1.00 (0.22-4.47)	0.99	0.85 (0.19-3.92)	0.84
Spontaneous MI	4 (1.0)	4 (1.4)	0.74 (0.19-2.95)	0.67	0.61 (0.15-2.44)	0.48
Ischemia-driven revascularization	59 (15.1)	23 (7.9)	1.99 (1.23-3.23)	0.004	1.90 (1.18-3.07)	0.008
All-cause death	10 (2.5)	10 (3.4)	0.75 (0.31-1.81)	0.52	0.70 (0.29-1.69)	0.43
Stroke	4 (1.0)	6 (2.1)	0.49 (0.14-1.73)	0.25	0.47 (0.13-1.67)	0.24

Values are n (%). MACE are defined as cardiac death, spontaneous MI, and ischemia-driven revascularization. IPTW variables included age, sex, diabetes, J-CTO score, lesion length, calcification, and post-PCI distal vessel RVD.

IPTW = inverse probability of treatment weighting; MACE = major adverse cardiac events; other abbreviations as in [Tables 1 and 2](#).

difference in the risk for MACE between the 2 groups within the first year, whereas the MB group exhibited a significantly elevated MACE risk after 1 year (HR: 2.22; 95% CI: 1.23-4.00; $P = 0.007$); similarly, the risk for ischemia-driven revascularization in the MB group was also significantly increased beyond 1 year (HR: 1.99; 95% CI: 1.10-3.62; $P = 0.02$) (Supplemental Figures 3C and 3D).

The results were consistent after IPTW analysis for 5-year MACE (adjusted absolute risk difference 8.07%; HR: 1.79; 95% CI: 1.13-2.82; $P = 0.01$) and ischemia-driven revascularization (adjusted absolute risk difference 8.38%; HR: 1.90; 95% CI: 1.18-3.07; $P = 0.008$). No significant differences were found in all-cause death, cardiac death, spontaneous MI, stroke, and bleeding events after IPTW adjustment ($P > 0.05$ for all) (Table 4).

MULTIVARIABLE ANALYSIS. Multivariable Cox regression analysis revealed that LAD CTO with an MB (HR: 1.81; 95% CI: 1.00-3.29) and CTO length (HR: 1.03; 95% CI: 1.00-1.06) were independent predictors of MACE (Supplemental Table 2). Analysis of the MB characteristics identified that stent extension to the MB was an independent predictor of MACE (HR: 1.91; 95% CI: 1.15-3.19; $P = 0.01$), while the systolic vessel area at the maximal compression site served as an independent protective factor against MACE (HR: 0.61; 95% CI: 0.41-0.91; $P = 0.02$). Additionally, total MB length, maximum MB thickness, diastolic vessel area at the maximal compression site, and plaque burden at the MLA site showed no significant association with MACE risk (Supplemental Table 3).

DISCUSSION

This large-scale, IVUS-based observational study demonstrated 3 principal findings: 1) concurrent MBs were present in 57.4% of patients with LAD CTOs on IVUS; 2) among these patients, nearly one-half had stents extending into the bridged segment; and 3) the presence of MBs was associated with poorer clinical outcomes, and those patients with stent extension into MBs experienced a worse long-term prognosis compared with patients in whom MBs were spared. These data support the clinical recommendation to avoid stent placement in the bridged segment whenever feasible.

An MB is a common coronary anatomical variant in the LAD territory, and its detection efficacy is significantly dependent on the diagnostic modality used. In the general population, the reported prevalence of MBs is 0.8% to 3.3% by conventional

coronary angiography, whereas IVUS-based studies have shown rates of 22% to 25.8%, underscoring the substantial influence of detection modality.^{4,18-21} Data in patients with CTOs, however, remain limited. Yamamoto et al⁴ reported that approximately 40% of patients with LAD CTOs are associated with MBs. The relatively high prevalence observed in the present study is broadly consistent with this finding and may reflect both the selected LAD CTO population and the greater sensitivity of IVUS compared with angiography for identifying MBs.^{4,21} These findings support the value of IVUS for accurate assessment of MBs, which may be important for avoiding stent implantation within the bridged segment.

This study also explored the association between different MB characteristics and the risk for MACE. The results showed no significant correlation between total MB length, maximum MB thickness, or MB compression ratio and MACE risk. This finding differs from some previous studies, which suggested that greater MB length and higher compression ratio might worsen prognosis by exacerbating myocardial ischemia. However, other studies have also failed to identify similar associations.⁶ We hypothesize that this discrepancy is related to the specific study population; this study was focused on LAD CTO patients, who inherently present with severe occlusive lesions. In such patients, prognosis may be influenced primarily by CTO lesion characteristics (such as lesion length and calcification) and the effectiveness of revascularization, while the impact of MB-specific features such as length and thickness on prognosis may be relatively diminished.

The most notable finding of this study is that LAD CTO patients with MBs had a poorer long-term prognosis, with a significantly higher 5-year rate of MACE. This outcome is attributable mainly to the markedly elevated rate of ischemia-driven revascularization in patients with stents extending into the MB segment. Furthermore, Kaplan-Meier curve analysis further clarified the stratified risk profile of this condition: the rate of ischemia-driven revascularization in MB patients with stents not extending into the MB was not significantly different from that of patients without MBs. Landmark analysis further showed that the elevated MACE risk in the MB group emerged after 1 year of follow-up, suggesting that the adverse prognostic impact of MB may be a long-term cumulative process. The underlying mechanism is multifactorial. First, systolic compression by MB causes periodic coronary flow restriction. Stent implantation disrupts the vessel's natural compression and alters shear stress, possibly triggering chronic myocardial ischemia and accelerating proximal

atherosclerosis.^{3,22-24} This progression may lead to in-stent restenosis or new lesions, increasing the need of repeat revascularization. Second, our study found that the subgroup of MB patients with stents extended into the MB had a higher MACE rate when dissection extended into the MB, which may be due to sustained systolic compression on the stented segment within the MB region, resulting in increased stent and vascular wall stress and subsequently elevated risks for dissection and restenosis.^{6,25}

The rate of stent extension into the MB segment in our cohort was 44.3%, validating the recent observation (48.7%) by Yamamoto et al.⁴ With a 3-fold larger sample size, our study confirms that this risk is not an artifact of small sample size. This indicates that this scenario is common during PCI for LAD CTOs and is difficult to completely avoid because of technical factors such as the close proximity of the lesion to the MB or dissection extension into the bridged segment. However, this strategy was significantly associated with adverse long-term outcomes. Several potential pathophysiological mechanisms have been proposed on the basis of previous clinical studies: the rigid stent disrupts the physiological contractility of the MB segment, and when coupled with the inherent cyclic compression from the overlying myocardium, this may lead to chronic vessel recoil or stent strut fracture, although the latter was not directly assessed in the present cohort, because routine imaging follow-up was not performed. These alterations collectively disturb local hemodynamic status, creating regions of low wall shear stress that promote neointimal hyperplasia and subsequent restenosis.^{23,24} This mechanism is supported by clinical studies from Xu et al²⁶ and Tsujita et al,²¹ both demonstrating that stent extension into the MB significantly increases the risks for restenosis and target lesion revascularization.

To reduce the risk for adverse events associated with stent extension into the MB, accurate assessment of coronary dissection before stenting is critical. IVUS indications for active intervention include media/adventitia-involving dissection affecting >50% of the vessel circumference, spiral or circumferential dissection in critical vascular segments, flow-limiting dissection with true luminal compression, and >5-mm edge dissection causing stent malapposition.^{27,28} Existing evidence indicates that stenting can be omitted in non-flow-limiting dissections with residual stenosis <30%. When dissection occurs adjacent to the MB during PCI, a systematic evaluation combining angiography and IVUS is mandatory to assess the extent of dissection, distal flow status, and MB involvement.

For non-flow-limiting dissections, deferred stenting is recommended; for flow-compromising cases, prolonged inflation with a noncompliant balloon should be adopted as the first-line strategy. Targeted spot stenting should be performed only if flow-limiting dissection persists, with every effort made to spare the MB segment.²⁹ Studies have shown that drug-coated balloons (DCBs) achieve long-term patency rates comparable with those of drug-eluting stents in CTO lesions and may therefore represent a stentless alternative for reducing MB-related adverse events when stent extension across the bridged segment would otherwise be required.^{27,30-32} Although ongoing prospective studies such as Co-CTO (Drug-Coated Balloon Coronary Angioplasty Versus Stenting for Treatment of Disease Adjacent to a Chronic Total Occlusion; [NCT04881812](#)),³³ REVERSE (Drug-Coated Balloon vs. Drug-Eluting Stent for Clinical Outcomes in Patients With Large Coronary Artery Disease; [NCT05846893](#)), IMAGINATION (Drug-Coated Balloon in Native Chronic Total Occlusion Percutaneous Coronary Intervention; [NCT06275386](#)), and DCB-CTO (Drug-Coated Balloon Versus Drug-Eluting Stent Angioplasty for Treatment of Chronic Total Occlusion; [NCT07369765](#)) are currently evaluating the clinical utility of DCB in CTO, evidence-based standardization for the specific subgroup of patients with concomitant myocardial bridging remains limited.

In summary, the recanalization of LAD CTOs with concomitant MB requires a carefully balance between acute results and the long-term risks associated with stenting the bridged segment. Our data support a precision PCI strategy in which “spot stenting” that spares the bridged segment is prioritized during LAD CTO intervention. When stenting across the MB cannot be avoided, DCBs may represents a viable alternative for reducing stent-related adverse events. Clinical decisions should be based on precise intracoronary imaging and functional evaluation, to develop individualized treatment strategies aimed at long-term outcomes.

STUDY LIMITATIONS. First, this study was a retrospective, single-center study that included only Chinese patients. It is subject to selection bias and may be affected by unmeasured variables. Whether the conclusions drawn from large cardiac centers can be generalized to other centers remains to be verified.

Second, we excluded the patients without IVUS imaging following CTO recanalization, and our findings cannot be extended to those angiography-guided cases, though an angiography-only approach

may theoretically result in more stents' extending into the bridged segments.

Third, after the initial PCI procedure, patients did not undergo routine coronary angiography or intravascular imaging follow-up. This prevented us from evaluating the association between stent-related complications (such as in-stent restenosis) and clinical outcomes.

Fourth, we evaluated MBs only in the LAD, limiting the applicability of the findings in other vessels, although MBs seldomly involve the right coronary artery or left circumflex artery.

Fifth, we exclusively enrolled patients with LAD CTO lesions. Accordingly, the findings cannot be extrapolated to patients without CTOs, and the association between stenting across MB segments and clinical outcomes in non-CTO patients requires further dedicated investigation. Future large-scale prospective randomized trials are warranted to validate the impact of MBs on the long-term prognosis for this complex anatomical subset.

CONCLUSIONS

Among patients with LAD CTOs undergoing IVUS-guided PCI, myocardial bridging is a frequent finding with worse long-term outcomes; the prognostic risk is driven primarily by stent extension into the bridged segment. Strategies that spare the bridged segment achieve long-term outcomes comparable with those in patients without MBs.

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PERSPECTIVES

WHAT IS KNOWN? MBs occur most frequently in the mid-LAD, and intravascular imaging substantially increases their detection. In non-CTO populations, stent implantation across an MB has been linked to higher rates of in-stent restenosis and target lesion revascularization, but the long-term prognostic impact of MBs and of stent extension into the bridged segment in patients undergoing PCI of LAD CTOs has not been well characterized.

WHAT IS NEW? IVUS identified MBs in 57.4% of patients undergoing PCI for LAD CTOs, and stent extension into the bridged segment occurred in 44.3% of MB cases. The presence of MBs was associated with worse 5-year outcomes, but this risk was driven mostly by stent extension into the bridged segment; when the MB was spared from stenting, long-term outcomes were comparable with those of patients without an MB.

WHAT IS NEXT? The optimal management of LAD CTOs with MBs remains unclear; further research is needed to refine MB diagnostics, validate long-term benefits of MB-sparing stenting, and standardize clinical care.

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KEY WORDS chronic total occlusions, left anterior descending, long-term outcome, myocardial bridge

APPENDIX For supplemental methods, tables, and figures, please see the online version of this paper.